E – COMMERCE ORDER MANAGEMENT SYSTEM (OMS)

Business Analysis Project Document

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1. Introduction

The rapid expansion of e-commerce has increased the complexity of managing customer orders efficiently. Businesses must handle multiple stages such as order placement, payment processing, shipment, and returns while maintaining accuracy and speed. In many cases, fragmented systems and manual processes create delays and errors in order handling. Customers also expect real-time updates and seamless experiences throughout the order lifecycle. Without a structured system, organizations struggle to maintain visibility and coordination across operations.

An effective solution is required to streamline these processes and improve efficiency. The E-commerce Order Management System (OMS) addresses these challenges by centralizing order-related activities. It enables better tracking, reduces manual effort, and enhances overall operational performance. This project focuses on analysing and designing such a system from a business perspective.

1.1 Document Purpose

This document provides a detailed business analysis for the proposed E-commerce Order Management System (OMS). It outlines the project context, problem statement, objectives, scope, and high-level requirements to improve order processing efficiency and customer experience.

1.2 Project Overview

The E-commerce Order Management System (OMS) is designed to streamline and centralize the end-to-end order lifecycle within an online retail environment. The system facilitates efficient handling of customer orders from initiation through fulfilment, delivery, and returns, ensuring seamless coordination between various stakeholders such as customers, administrators, vendors, and delivery partners.

The proposed OMS will serve as a unified platform that enables real-time tracking of orders, automated status updates, and improved visibility across all stages of the order process. By reducing dependency on manual processes and fragmented systems, the solution aims to enhance operational efficiency, minimize errors, and provide a consistent and transparent experience to end users.

This project focuses on analysing business needs, identifying process gaps, and defining structured requirements to support the development of a scalable and efficient order management system aligned with modern e-commerce practices.

1.3 Problem Statement

E- commerce businesses often struggle with inefficient order processing, lack of real-time tracking, and poor inventory coordination. These challenges lead to delays, errors, and reduced customer satisfaction.

Problem Area	Current Challenge	Business Impact
Order Tracking	No real-time tracking available	Customer dissatisfaction, high support calls
Order Status Updates	Manual email-based updates by admin	Delayed and inconsistent communication
Order Cancellation	Customers must call or email to cancel	Slow processing, no immediate confirmation
Return and Refund	Manual paperwork-based process	Lengthy resolution, poor visibility
Admin Monitoring	Spreadsheet-based order management	Error-prone, lacks real-time visibility

1.4 Business Objectives

- Improve overall efficiency of the order management process across the organization
- Enhance customer satisfaction through real-time order visibility and timely updates
- Reduce manual intervention and operational errors in order processing
- Ensure better coordination and communication between internal and external stakeholders
- Increase process transparency and accountability across the order lifecycle
- Support business growth by enabling a scalable and structured order management framework

Objective ID	Business Objective	Success Metric
BO-01	Improve order processing efficiency	40% reduction in processing time
BO-02	Enhance customer satisfaction	Improved customer satisfaction score
BO-03	Reduce manual intervention	60% reduction in manual order handling
BO-04	Improve stakeholder coordination	Faster communication across all parties
BO-05	Increase process transparency	100% real-time order status visibility
BO-06	Support business scalability	System supports 30% annual order growth

1.5 Project Objectives

- Design and document a comprehensive Order Management System (OMS) to support end-to-end order processing
- Enable customers to place, track, and manage orders efficiently through the system
- Provide administrators with a centralized dashboard to monitor and manage order activities
- Implement functionalities for order status updates, cancellations, and returns management
- Define clear functional and non-functional requirements to guide system development
- Establish structured workflows to improve accuracy and reduce processing time
- Deliver a well-documented Business Analysis solution including BRD, FRD, and user stories

1.6 Glossary of Terms

- OMS - Order Management System — a centralized platform to manage end-to-end order processing
- BRD - Business Requirements Document — captures high-level business needs
- FRD - Functional Requirements Document — captures detailed system functionalities
- RTM - Requirements Traceability Matrix — links requirements to user stories and test cases
- SLA - Service Level Agreement — defines expected system performance standards
- RTO - Recovery Time Objective — maximum acceptable downtime after a failure
- RPO - Recovery Point Objective — maximum acceptable data loss after a failure
- UAT - User Acceptance Testing — validation by end users before system go-live
- MoSCoW - Prioritization method: Must Have, Should Have, Could Have, Won't Have
- RACI - Responsibility matrix: Responsible, Accountable, Consulted, Informed
- SKU - Stock Keeping Unit — unique identifier for each product
- API - Application Programming Interface — enables communication between systems

2. Project Scope

The scope of this project defines the boundaries and functionalities of the proposed E-commerce Order Management System (OMS). It outlines the features and processes that are included within the system, as well as those that are explicitly excluded to maintain clarity and focus.

2.1 In Scope

#	Feature	Description
1	User Registration and Authentication	Customers can register, log in, and log out of the system securely
2	Product Browsing and Selection	Customers can browse the product catalogue and select items for purchase
3	Order Placement and Confirmation	Customers can place orders and receive an order confirmation with a unique Order ID
4	Payment Status Tracking	System tracks and displays payment status as Successful, Pending, or Failed
5	Real-time Order Tracking	Customers can view live order status updates throughout the order lifecycle
6	Order Lifecycle Management	System manages order stages from Placed → Processed → Shipped → Delivered
7	Order Cancellation	Customers can cancel orders before they are shipped
8	Returns and Refund Initiation	Customers can initiate return and refund requests for eligible delivered orders
9	Administrative Dashboard	Administrators can view, manage, and update all order activities from a central dashboard
10	System Notifications	Customers receive system notifications for key order status updates

2.2 Out of Scope

#	Excluded Feature	Reason for Exclusion
1	Inventory and Warehouse Management	Falls under a separate system scope outside OMS boundaries
2	Third-party Delivery Logistics Integration	Considered at a high level only; deep integration is excluded
3	Supplier and Vendor Management System	Outside the current project boundary
4	Advanced Analytics and Reporting Dashboards	Planned for a future project phase
5	AI-based Product Recommendations	Future phase consideration; not part of core OMS functionality
6	Payment Gateway Internal Processing	Responsibility of the external payment gateway provider
7	Mobile Application Development	Focus is on system functionality, not platform-specific implementation

2.3 Scope Limitations

#	Limitation	Impact
1	This is a conceptual Business Analysis initiative	No actual system development or deployment is involved
2	External system integrations are considered at a high level only	Payment gateways and delivery partners are assumed, not deeply integrated
3	Data used within the project is hypothetical	All data is intended for analysis purposes only and does not represent real business data

3. Stakeholders

Stakeholders are individuals or groups who are actively involved in the project or whose interests may be positively or negatively impacted by the implementation of the E-commerce Order Management System (OMS). Identifying stakeholders is essential to ensure that business needs and expectations are properly captured and addressed.

3.1 Primary Stakeholders

These stakeholders directly interact with the system on a regular basis:

- **Customers (End Users)**
  Responsible for browsing products, placing orders, making payments, tracking orders, and initiating returns or cancellations.
- **System Administrators**
  Manage and monitor the overall system, including order processing, status updates, and user management.
- **Vendors / Sellers**
  Responsible for managing product availability, order fulfilment, and dispatching orders.

3.2 Secondary Stakeholders

These stakeholders support the system indirectly:

- **Delivery Partners / Logistics Teams**
  Handle the shipment and delivery of orders to customers and provide delivery status updates.
- **Payment Gateway Providers**
  Facilitate secure online payment processing and transaction status updates.

3.3 Business Stakeholders

These stakeholders define business goals and oversee system success:

- **Business Owners / E-commerce Management Team**
  Define business requirements, monitor performance, and ensure the system aligns with organizational goals.

- **Business Analyst (BA)**
  Acts as a bridge between business stakeholders and technical teams by gathering, analysing, and documenting requirements.

3.4 Stakeholder Expectations

- Seamless and efficient order processing system
- Real-time visibility of order status
- Reduced operational delays and errors
- Improved customer experience and satisfaction
- Transparent communication between all parties involved in the order lifecycle

3.5 RACI Matrix

The RACI Matrix defines the roles and responsibilities of each stakeholder across key order management activities.

Activity	Customer	Admin	Vendor	Delivery Partner	Business Analyst	Business Owner
Place Order	R	I	I	—	—	—
Confirm Payment	I	A	—	—	—	—
Process Order	I	R	A	—	—	—
Pack Order	—	I	R	—	—	—
Ship Order	I	I	A	R	—	—
Deliver Order	I	I	—	R	—	—
Update Order Status	—	R	C	C	—	—
Cancel Order	R	A	I	—	—	—
Initiate Return	R	A	I	—	—	—
Requirements Sign-Off	—	C	—	—	R	A
Monitor Order Activity	—	R	—	—	—	A

R = *Responsible* | A = *Accountable* | C = *Consulted* | I = *Informed*

4. Assumptions

Assumptions are conditions considered to be true for the successful execution of the E-commerce Order Management System (OMS) project. These assumptions are made to define a stable baseline for analysis and documentation.

#	Assumption	Impact if Not True
1	Users have stable internet and basic digital literacy	System adoption may be affected
2	Payment gateways function correctly and securely	Orders cannot be processed
3	Delivery partners provide accurate status updates	Tracking information will be inaccurate
4	Stakeholders actively participate in requirement sessions	Requirements may be incomplete
5	Data used for analysis is accurate and consistent	Analysis outcomes may be unreliable

5. Constraints

Constraints are limitations or restrictions that may impact the design, development, or implementation of the E-commerce Order Management System (OMS). These constraints must be considered while defining requirements.

#	Constraint	Impact
1	Limited project timeline	Scope must remain focused
2	No actual system development involved	Analysis is conceptual only
3	External integrations are assumed only	High-level analysis only
4	Scope restricted to order management	Full e-commerce platform excluded
5	Limited access to real-time business data	Hypothetical data used

6. Business Requirements (MoSCoW TABLE)

Business Requirements define the high-level needs of the organization that the E-commerce Order Management System (OMS) must fulfil. These requirements focus on *what the business expects the system to achieve* rather than how it will be implemented.

The goal of the OMS is to improve order processing efficiency, enhance customer experience, and ensure smooth coordination between all stakeholders involved in the order lifecycle.

6.1 Functional Business Requirements

Requirement ID	Requirement Description	Priority (MoSCoW)
BR-01	The system shall allow customers to register and log in securely	Must Have
BR-02	The system shall allow customers to browse and select products	Must Have
BR-03	The system shall generate a unique Order ID for each confirmed order	Must Have
BR-04	The system shall provide real-time order status updates	Must Have
BR-05	The system shall allow customers to cancel orders before shipment	Must Have
BR-06	The system shall support return and refund initiation for eligible orders	Should Have
BR-07	The system shall allow administrators to manage and update order statuses	Must Have
BR-08	The system shall notify customers about key order updates	Should Have
BR-09	The system shall provide an admin dashboard for order activity overview	Should Have
BR-10	The system shall support multiple payment methods	Could Have

6.2 Non-Functional Business Requirements

Requirement ID	Requirement Description	Priority (MoSCoW)
NFR-01	System page load time shall not exceed 3 seconds under normal load	Must Have
NFR-02	System shall support up to 10,000 concurrent users without degradation	Must Have
NFR-03	System uptime shall be maintained at 99.9% as per SLA	Must Have
NFR-04	All sensitive user data shall be encrypted using industry-standard protocols	Must Have
NFR-05	System shall recover within 15 minutes of failure (RTO ≤ 15 mins)	Should Have
NFR-06	Data loss in case of failure shall not exceed 1 hour of transactions (RPO ≤ 1 hr)	Should Have
NFR-07	System shall provide a user-friendly and consistent interface	Must Have
NFR-08	System shall be scalable to accommodate future business growth	Should Have

6.3 MoSCoW Prioritization Summary

Priority	Requirements
Must Have	User login, order placement, order tracking, order ID generation, admin status update, data encryption, system uptime
Should Have	Return and refund, notifications, admin dashboard, scalability, RTO/RPO compliance
Could Have	Multiple payment methods, advanced order filters, email notifications
Won't Have (this release)	AI recommendations, inventory management, mobile app, third-party logistics integration

7. AS-IS vs TO-BE Process Analysis

This section presents the current state of order management challenges and the proposed future state that the OMS will deliver.

Process Area	AS-IS (Current State)	Pain Point	TO-BE (Future State)
Order Tracking	Customers call support to get order updates	Time-consuming, inaccurate	Real-time tracking visible in customer portal
Order Status Updates	Email-based manual updates sent by admin	Delayed, inconsistent	Automated instant notifications at every stage
Order Cancellation	Customer raises request via phone or email	Slow processing, no confirmation	Self-service cancellation through the system
Return & Refund	Manual paperwork and approvals required	Lengthy process, poor visibility	System-driven return workflow with status tracking
Admin Monitoring	Spreadsheet-based order tracking	Error-prone, lacks visibility	Centralized dashboard with real-time data
Payment Status	Manually verified by admin	Delays in order processing	Automated payment validation and confirmation
Stakeholder Communication	Phone and email-based	Fragmented, slow	System notifications to all relevant stakeholders

8. Process Flow (Order Lifecycle)

The Process Flow defines the end-to-end sequence of activities involved in the E-commerce Order Management System (OMS). It represents how an order moves through different stages from initiation to completion, ensuring clarity in business operations and stakeholder responsibilities.

8.1 Order Lifecycle Flow

The typical order process in the system follows the steps below:

1. Customer logs into the system
2. Customer browses and selects products
3. Customer adds products to cart and places an order
4. System generates a unique Order ID
5. Payment is processed through the payment gateway
6. Order confirmation is generated upon successful payment
7. Order is processed by the system/admin
8. Order is packed by vendor/seller
9. Order is shipped by delivery partner
10. Customer receives the order
11. Order status is updated to “Delivered”
12. Customer may initiate return or refund (if applicable)

8.2 Order Status Flow

The system maintains clear tracking of order status at each stage:

- Order Placed
- Payment Confirmed
- Order Processed
- Order Packed
- Order Shipped
- Out for Delivery

- Delivered
- Returned / Cancelled (if applicable)

8.3 Process Description

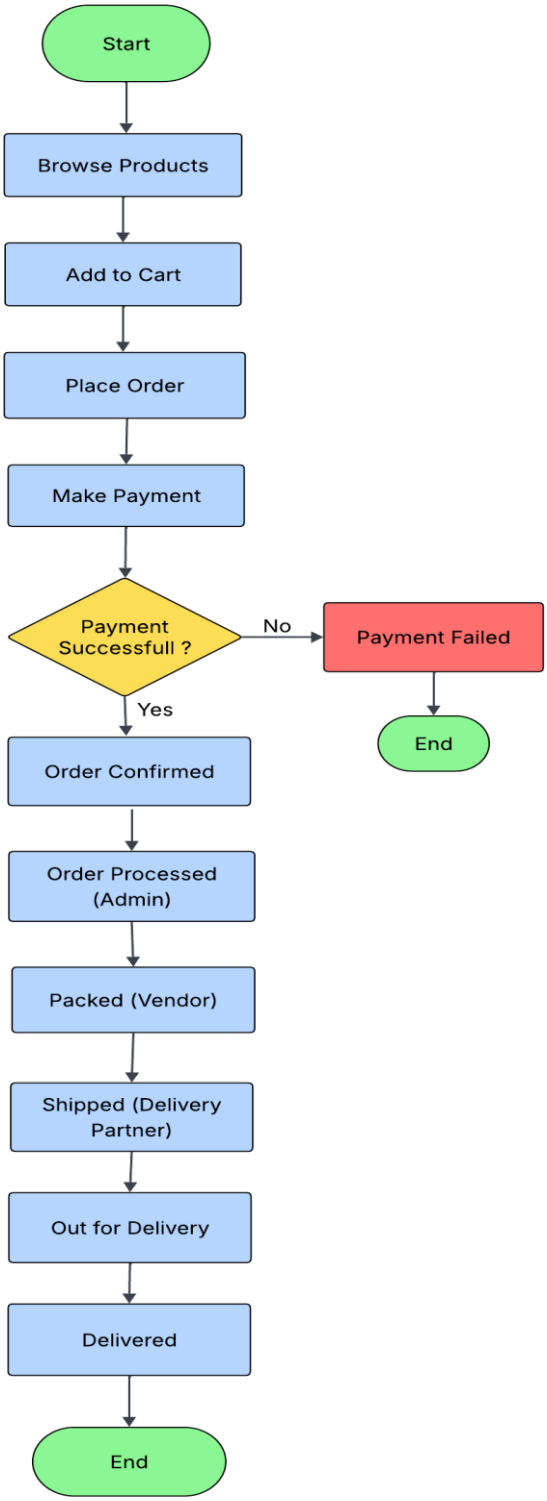
The OMS ensures real-time updates at each stage of the order lifecycle. Once a customer places an order, the system validates payment and generates an order confirmation. The order is then processed internally, followed by packaging and shipment through delivery partners. Each stage update is reflected in the system to provide transparency to both customers and administrators.

In case of cancellation or return requests, the system allows controlled reversal of the order process based on business rules and eligibility criteria.

8.4 Business Value of Process Flow

- Improves transparency in order tracking
- Reduces communication gaps between stakeholders
- Ensures structured and standardized order handling
- Enhances customer trust through real-time updates
- Helps identify bottlenecks in the order lifecycle

8.5 Process Flow Diagram



"This flow represents the end-to-end order lifecycle from order placement to delivery and completion."

9. Functional Requirements

Functional Requirements define the detailed behaviour and functionalities that the E-commerce Order Management System (OMS) must perform to support business processes.

FR ID	Functional Requirement	Priority
FR-01	System shall allow users to register with valid personal details	Must Have
FR-02	System shall authenticate users through secure login credentials	Must Have
FR-03	System shall allow users to update their profile information	Should Have
FR-04	System shall allow users to reset their password securely	Must Have
FR-05	System shall allow users to browse available products	Must Have
FR-06	System shall allow users to add products to the cart	Must Have
FR-07	System shall allow users to place an order successfully	Must Have
FR-08	System shall generate a unique Order ID for each transaction	Must Have
FR-09	System shall provide real-time order status updates	Must Have
FR-10	System shall update order status at each stage of the lifecycle	Must Have
FR-11	System shall notify users of key order status changes	Should Have
FR-12	System shall support payment processing through integrated gateway	Must Have
FR-13	System shall allow users to cancel orders before shipment	Must Have
FR-14	System shall validate cancellation eligibility based on order status	Must Have
FR-15	System shall allow users to initiate return requests after delivery	Should Have
FR-16	System shall process refund requests based on business rules	Should Have
FR-17	System shall allow administrators to view and manage all orders	Must Have

9.1 Business Value of Functional Requirements

- Ensures structured and automated order processing
- Reduces dependency on manual intervention
- Improves accuracy in order tracking and management
- Enhances customer experience through timely updates
- Supports efficient coordination between stakeholders

10. Non-Functional Requirements

Non-Functional Requirements define the quality attributes, system performance standards, and operational constraints of the E-commerce Order Management System (OMS). These requirements ensure that the system performs efficiently, securely, and reliably under expected conditions.

10.1 Performance Requirements

Requirement ID	Requirement	Measurable Standard
NFR-P01	Page load time	≤ 3 seconds under normal operating conditions
NFR-P02	Order details retrieval	≤ 2 seconds response time
NFR-P03	Concurrent user support	Minimum 10,000 simultaneous users without degradation
NFR-P04	Order status update processing	Real-time updates within 5 seconds of status change

10.2 Security Requirements

Requirement ID	Requirement	Measurable Standard
NFR-S01	User authentication	Secure login with encrypted password storage (equivalent)
NFR-S02	Data encryption	All sensitive data encrypted using AES-256 or equivalent
NFR-S03	Access control	Role-based access control (RBAC) restricting admin functions
NFR-S04	Data protection compliance	Adherence to basic data protection and privacy standards

10.3 Availability and Reliability

Requirement ID	Requirement	Measurable Standard
NFR-A01	System uptime	99.9% availability (SLA) — maximum 8.7 hours downtime per year
NFR-A02	Recovery Time Objective	System restored within 15 minutes of failure (RTO ≤ 15 mins)
NFR-A03	Recovery Point Objective	Maximum 1 hour of data loss acceptable (RPO ≤ 1 hour)
NFR-A04	Order data reliability	100% accuracy in order status updates across all modules

10.4 Usability Requirements

Requirement ID	Requirement	Measurable Standard
NFR-U01	Interface usability	New users should complete order placement within 5 minutes
NFR-U02	Navigation consistency	Uniform layout and design across all system pages
NFR-U03	Error messaging	Clear error messages displayed within 2 seconds of invalid action

10.5 Scalability Requirements

Requirement ID	Requirement	Measurable Standard
NFR-SC01	User scalability	System supports 30% annual growth in user base without redesign
NFR-SC02	Order volume scalability	Handles up to 1,00,000 orders per day during peak season

10.6 Data Integrity

Requirement ID	Requirement	Measurable Standard
NFR-D01	Data consistency	Order data synchronized across all modules in real time
NFR-D02	Duplicate prevention	Zero duplicate Order IDs across all transactions
NFR-D03	Transaction accuracy	100% data accuracy maintained during payment and order processing

11. User Stories with Acceptance Criteria

User Stories represent system requirements from the end-user perspective in an Agile format. They help in understanding user needs and expected business value in a simple and structured way.

Each user story follows the standard Agile format:
As a, I want, so that.

11.1 Customer User Stories

User Story -01: User Registration and Login

As a customer, I want to register and log in to the system so that I can access my account securely.

Acceptance Criteria	
AC-01	Given the customer provides valid details, when they register, then account is created successfully
AC-02	Given the customer enters correct credentials, when they log in, then they are redirected to the home page
AC-03	Given the customer enters incorrect credentials, when they attempt login, then an error message is displayed
AC-04	Given the customer is inactive for 15 minutes, when the session expires, then they are automatically logged out

User Story -02: Browse Products

As a customer, I want to browse products so that I can select items for purchase.

Acceptance Criteria	
AC-01	Given the customer is logged in, when they navigate to the product catalogue, then all available products are displayed
AC-02	Given a product is out of stock, when the customer views it, then an "Out of Stock" label is displayed
AC-03	Given the customer searches for a product, when results load, then relevant products appear within 2 seconds

User Story -03: Place Order

As a customer, I want to place an order so that I can purchase selected products.

Acceptance Criteria	
AC-01	Given the customer completes payment, when the order is confirmed, then a unique Order ID is generated
AC-02	Given payment fails, when the customer is notified, then the order is not placed and they are prompted to retry
AC-03	Given the order is placed successfully, when confirmation is generated, then the customer receives a system notification

User Story -04: Track Order

As a customer, I want to track my order so that I can know the delivery status in real time.

Acceptance Criteria	
AC-01	Given the order is placed, when the customer navigates to "My Orders", then current order status is displayed
AC-02	Given the order is shipped, when the customer views tracking, then tracking ID and estimated delivery date are shown
AC-03	Given the order is delivered, when the customer checks status, then "Delivered" is shown with date and timestamp

User Story -05: Cancel Order

As a customer, I want to cancel my order so that I can avoid unwanted purchases.

Acceptance Criteria	
AC-01	Given the order is not yet shipped, when the customer clicks "Cancel Order", then the order status is updated to "Cancelled"
AC-02	Given the order has already been shipped, when the customer attempts cancellation, then the system displays "Cancellation not available"
AC-03	Given the order is cancelled, when the process is complete, then the customer receives a cancellation confirmation notification

User Story -06: Initiate Return

As a customer, I want to initiate a return so that I can return faulty or unwanted products.

Acceptance Criteria	
AC-01	Given the order is delivered, when the customer selects "Return Order", then a return request form is displayed
AC-02	Given the return request is submitted, when it is validated, then status is updated to "Return Initiated"
AC-03	Given the return is not eligible (e.g., return window expired), when the customer attempts return, then an appropriate message is shown

11.2 Admin User Stories

User Story -07: View All Orders

As an admin, I want to view all customer orders so that I can monitor system activity.

Acceptance Criteria	
AC-01	Given the admin is logged in, when they navigate to the dashboard, then all orders with statuses are displayed
AC-02	Given the admin applies a filter, when results load, then only orders matching the filter criteria are shown

User Story -08: Update Order Status

As an admin, I want to update order status so that customers receive accurate tracking information.

Acceptance Criteria	
AC-01	Given the admin selects an order, when they update the status, then the new status is reflected immediately in the system
AC-02	Given the status is updated, when the change is saved, then the customer receives an automated notification

User Story -09: Manage Cancellations and Returns

As an admin, I want to manage cancellations and returns so that business rules are followed.

Acceptance Criteria	
AC-01	Given a cancellation request is raised, when the admin reviews it, then they can approve or reject based on eligibility
AC-02	Given a return is approved, when the admin confirms it, then the order status updates to "Return Approved"

11.3 Business Value of User Stories

- Provides clear understanding of user needs and expectations
- Helps in breaking down requirements into actionable tasks for development
- Improves communication between business and technical teams
- Supports Agile development and sprint planning
- Ensures user-centric system design and implementation

12. Use Cases

Use Cases describe the interaction between users (actors) and the system to achieve specific goals. They provide a detailed view of how different functionalities of the E-commerce Order Management System (OMS) are executed in real-world scenarios.

12.1 Use Case: Place Order

Field	Details
Use Case ID	UC-01
Use Case Name	Place Order
Actor	Customer
Description	Customer places an order for selected products
Preconditions	Customer is logged in; products are available in catalogue
Postconditions	Order is successfully created with a unique Order ID

Main Flow:

1. Customer selects products and adds to cart
2. Customer proceeds to checkout
3. System calculates total order amount
4. Customer selects payment method and completes payment
5. System validates payment confirmation
6. System generates unique Order ID
7. Order confirmation is displayed to customer
8. System notification is sent to customer

Alternate Flow — Payment Failure:

- Step 5a: Payment gateway returns failure response
- System displays "Payment Failed" message
- Customer is prompted to retry or use alternate payment method
- Order remains in "Pending Payment" status

Exception Flow — Session Timeout:

- If customer is inactive for 15 minutes during checkout
- System automatically logs out the customer
- Cart items are saved for the next login session

12.2 Use Case: Track Order

Field	Details
Use Case ID	UC-02
Use Case Name	Track Order
Actor	Customer
Description	Customer views real-time status of their placed order
Preconditions	Order has been placed successfully
Postconditions	Customer is informed of current order status

Main Flow:

1. Customer logs into system
2. Customer navigates to "My Orders" section
3. System retrieves and displays order list
4. Customer selects specific order
5. System displays current order stage with timestamp

Alternate Flow — No Orders Found:

- System displays "No orders found" message
- Customer is prompted to start shopping

12.3 Use Case: Cancel Order

Field	Details
Use Case ID	UC-03
Use Case Name	Cancel Order
Actor	Customer
Description	Customer cancels an order that has not yet been shipped
Preconditions	Order exists and has not been shipped
Postconditions	Order status updated to "Cancelled"

Main Flow:

1. Customer navigates to "My Orders"
2. Customer selects the order to cancel
3. Customer clicks "Cancel Order"
4. System validates cancellation eligibility
5. System cancels the order and updates status to "Cancelled"
6. Customer receives cancellation notification

Alternate Flow — Order Already Shipped:

- System detects order is in "Shipped" or later status
- System displays "Cancellation not available at this stage"
- Customer is informed to raise a return request after delivery

12.4 Use Case: Process Order (Admin)

Field	Details
Use Case ID	UC-04
Use Case Name	Process Order
Actor	Admin
Description	Admin reviews and processes confirmed orders
Preconditions	Order is placed and payment is confirmed
Postconditions	Order status updated to "Processed" and moves to packaging stage

Main Flow:

1. Admin logs into system
2. Admin navigates to pending orders on dashboard
3. Admin reviews order details
4. Admin processes the order
5. System updates order status to "Processed"
6. Order is passed to vendor for packaging

Exception Flow — Incomplete Order Information:

- Admin identifies missing or incorrect order information
- Admin flags the order and contacts customer for clarification
- Order remains in "On Hold" status until resolved

12.5 Business Value of Use Cases

- Provides clear step-by-step system interaction view
- Helps bridge gap between business requirements and technical implementation
- Improves clarity for developers and testers
- Supports system design and test case creation
- Enhances understanding of real-world system behaviour

13. Risk Register

Risk ID	Risk Description	Likelihood	Impact	Risk Level	Mitigation Strategy
R-01	Payment gateway downtime causing failed transactions	Medium	High	High	Integrate backup payment provider; retry mechanism
R-02	Data breach or unauthorized access	Low	High	High	Implement encryption, RBAC, and security audits
R-03	High traffic causing system performance degradation	Medium	High	High	Load balancing and auto-scaling infrastructure
R-04	Incorrect order status updates causing customer dissatisfaction	Medium	Medium	Medium	Automated validation rules and admin review process
R-05	Delivery partner API failure affecting tracking updates	Medium	High	High	Manual override option for admin; fallback notifications
R-06	Scope creep extending project timeline	High	Medium	High	Strict change control process and scope sign-off
R-07	Stakeholder unavailability during requirement validation	Low	Medium	Low	Schedule reviews in advance; document all approvals
R-08	Data loss due to system failure	Low	High	High	Regular data backups; RTO and RPO compliance

14. Requirements Traceability Matrix (RTM)

Requirement ID	Business Requirement	Functional Requirement	User Story	Use Case	Test Case
BR-01	User registration and login	FR-01: System shall allow users to register and authenticate	US-01	—	TC-001
BR-02	Product browsing and order placement	FR-05: System shall allow users to browse and place orders	US-02, US-03	UC-01	TC-002
BR-03	Unique Order ID generation	FR-08: System shall generate unique Order ID per transaction	US-03	UC-01	TC-003
BR-04	Real-time order tracking	FR-09: System shall provide real-time status updates	US-04	UC-02	TC-004
BR-05	Order cancellation	FR-13: System shall allow cancellation before shipment	US-05	UC-03	TC-005
BR-06	Return and refund initiation	FR-15: System shall allow return requests after delivery	US-06	—	TC-006
BR-07	Admin order management	FR-17: Admin shall update and monitor all orders	US-07, US-08, US-09	UC-04	TC-007
BR-08	Customer notifications	FR-11: System shall notify users of order status changes	US-03, US-04, US-05	UC-01, UC-02	TC-008

15. UAT Test Scenarios

Test Case ID	Linked User Story	Test Scenario	Steps	Expected Result
TC-001	US-01	Valid user registration	Enter valid details → Submit → Check confirmation	Account created successfully
TC-002	US-01	Invalid login attempt	Enter wrong password → Click Login	Error message displayed
TC-003	US-03	Successful order placement	Add to cart → Checkout → Pay → Confirm	Unique Order ID generated
TC-004	US-03	Payment failure during checkout	Enter invalid card → Submit payment	Error shown; order not placed
TC-005	US-04	Order tracking after shipment	Login → My Orders → Select order	Tracking ID and delivery date shown
TC-006	US-05	Cancel order before shipment	Select order → Cancel → Confirm	Status updated to "Cancelled"
TC-007	US-05	Cancel order after shipment	Select shipped order → Attempt cancel	"Cancellation not available" message shown
TC-008	US-06	Return request after delivery	Select delivered order → Initiate return → Submit	Status updated to "Return Initiated"
TC-009	US-08	Admin updates order status	Admin login → Select order → Update status	Status updated; customer notified
TC-010	US-09	Admin approves return request	Admin reviews return → Approves	Status updated to "Return Approved"

16. Tools Used

This section outlines the tools and technologies used during the Business Analysis documentation of the E-commerce Order Management System (OMS) project. These tools support requirement gathering, documentation, process modelling, and visualization.

16.1 Documentation Tools

- **Microsoft Word**
  Used for preparing the Business Analysis document including BRD, FRD, use cases, and project documentation in a structured format.

16.2 Requirement & Project Management Tools

- **Jira**
  Used for creating and managing user stories, tracking requirements, and simulating Agile sprint planning.
- **Confluence**
  Used for documenting and sharing project requirements, BRD, FRD, and stakeholder information in a centralized workspace.

16.3 Process Modelling Tools

- **Lucid chart**
  Used to create process flow diagrams and visualize the end-to-end order lifecycle of the system.
- **Microsoft Visio**
  Used for designing detailed flowcharts and business process diagrams.

16.4 Agile Methodology

- **Scrum Framework**
  Used to structure user stories, prioritize requirements, and simulate sprint-based development cycles.
- **Agile Practices**
  Applied for breaking down requirements into smaller user stories and focusing on iterative development.

16.5 Business Analysis Skills Applied

- Requirement Gathering
- Stakeholder Analysis
- Business Process Modelling
- Problem Solving
- Use Case Design
- User Story Creation

16.6 Business Value of Tools

- Improves clarity and structure of documentation
- Enhances collaboration between business and technical teams
- Supports visualization of complex processes
- Enables efficient requirement tracking and management
- Aligns project with real-world industry BA practices

17. Conclusion

The E-commerce Order Management System (OMS) project provides a structured business analysis of an end-to-end order processing solution in an e-commerce environment. The project successfully identifies key business challenges such as inefficient order tracking, lack of process transparency, and dependency on manual operations.

Through detailed documentation including BRD, FRD, use cases, user stories, and process flows, this project defines a clear and organized framework for managing the complete order lifecycle—from order placement to delivery and returns.

The proposed system aims to improve operational efficiency, enhance customer experience, and ensure better coordination among all stakeholders including customers, administrators, vendors, and delivery partners.

By applying Business Analysis techniques such as requirement gathering, stakeholder analysis, and process modelling using tools like Lucid chart and Jira, this project demonstrates a practical understanding of real-world BA responsibilities.

Overall, this OMS project reflects a comprehensive approach to solving business problems through structured analysis and aligns with industry-standard Business Analysis practices.

"As part of this project deliverable, separate Business Requirements Document (BRD) and Functional Requirements Document (FRD) have been prepared to support the detailed analysis and documentation of the proposed system."